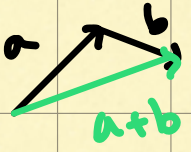
- vector addition = tip to tail



$\begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} a+c \\ b+d \end{bmatrix}$

- scalar multip = $s\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} sx \\ sy \end{bmatrix}$

- vector = movement / step in space
- like when you add numbers on a number line

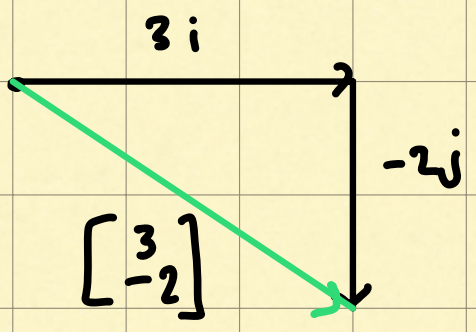
- language to describe space & manipulation of space.

#2: linear combinations, span, & basis vectors

- vector coordinates help this back & forth b/w pairs of numbers & 2D coordinates
- another way to think ab coordinates
- think ab each coordinate as a scalar
- think ab how each one stretches & squishes vectors
- Unit vector in x direction, $\hat{i}$
  " y " , $\hat{j}$

- x coordinate of a vector is a scalar stretching $\hat{i}$
- y " $\hat{j}$

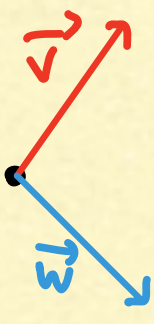
- ex $\begin{bmatrix} 3 \\ -2 \end{bmatrix} = (3)\hat{i} - 2\hat{j}$



- $\hat{i}$ & $\hat{j}$ are basis vectors of xy coordinates

- when you think of coordinates as scalars...
- basis vectors are what coordinates actually scale!

- what if we chose different basis vectors?
- you could have gotten a new coordinate system



- what are all the vectors you can get by choosing two scalars using each one to scale one of the vectors then adding together

what you get...

- which 2D vectors can be reached by altering choice of scalars?

You can reach every possible 2D vector

- a new pair of basis vectors, still gave a valid way to go back & forth b/w pairs of numbers & 2D vectors

- but association is different than what you get when u use standard basis vectors

- any time we describe vectors numerically, depends on choice of what basis vectors you chose

- linear combination = taking two vectors & scaling them

- The set of all vectors you can reach w/ linear combination of a given pair of vectors, is the **span**

- Span of most pairs of 2D vectors, is all 2D vectors, unless
  - → vectors are zero (stuck @ origin)
  - → vectors face same direction (then span is all vectors in the same direction, the line ...)

- span of 2 vectors = all vectors you can reach using only
  - vector addition
  - scalar multip

- vectors @ points = tail @ origin
- all 2D vectors all at once = infinite sheet of 2D space

- collection of vectors = pairs
- 1 vector = arrow

- span of 2 3D vectors?
- → is it incomplete? do u need 3 vectors?

- their span is all linear combinations of those two vectors

- the span of 2 vectors in 3D space is a 2D plane. (the set of all 3D vectors whose tip sits on that flat sheet)

- the span of 3 vectors in 3D
  → if third vector in span of 1st 2, if it's in that plane already, span

does not change. Just like when you were stuck on a line in 2D when both vectors lined same way

→ if 3rd vector outside span of $\vec{v}_1$ & $\vec{v}_2$, now you can reach all of 3D space

- you'd use all 3 dimensions to get all 3D space.

- when you have that redundant vector & not adding anything to our span

  2D: 1 vector in same direction as other

  3D: 1 vector in span of other two

- when I have multiple vectors & you can remove one w/out redefining the span... they are linearly dependent

- 1 vector can be expressed as linear combination of the other(s).

since it's already in the span of the others.

- if each vector adds new dimension to span, they are linearly independent & cannot be expressed as linear combinations of each other.

- puzzle

basis of a vector space is set of linearly independent vectors that span the full space.

makes sense bc the vectors you can use to reach every vector in the space, are the ones you stretch/squeeze w/ lin comb. = the basis vectors.

to prove set of vectors is a basis for a vector space, show that the set linearly independent $\Leftrightarrow$ spans the space

ex: determine if $v_1, v_2, v_3 = \begin{bmatrix}1\\2\\1\end{bmatrix}\begin{bmatrix}1\\0\\2\end{bmatrix}\begin{bmatrix}1\\1\\0\end{bmatrix}$ form a basis for R^3

$\rightarrow$ put into a matrix, compute determinant or reduce to RREF.

if $\det() \neq 0$, the set is linearly independent $\Leftrightarrow$ spans R^3